SQL Server Installation Checklist (Jonathan Kehayias (from SQLskills))

Source: https://www.sqlskills.com/blogs/jonathan/sql-server-installation-checklist/

Key steps, goals, and best practices for installing SQL Server correctly in production environments.

This checklist covers three major phases:
✅ **Windows OS Installation**
✅ **SQL Server Pre-Installation**
✅ **SQL Server Installation + Post-Installation Configuration**

1. Windows OS Installation (Foundation Setup)

System & Security Setup

1. **Partition Alignment:** Ensure disk partitions are aligned for optimal SQL IO.
2. **Hyper-Threading:** Enabled in BIOS to help CPU efficiency.
3. **Drive Layout:**
    - OS and applications on RAID1 with **NTFS default allocation size**.
    - Install OS on **C: drive**.
4. **Admin Rights:** Add **Domain Admins** to local admin group.

Security Policies

5. **Account Policies:**
    - Strong password requirements (length, complexity).
    - Account lockout on invalid login attempts.
6. **Audit Policies:** Enable auditing for logon and policy changes.
7. **Interactive Logon Settings:** Do not display last user name; show legal message.

Drive & Network Settings

8. **Remove Everyone** from non-C drives for security.
9. **Install apps** to **D: drive** to keep OS separate.
10. **Windows Updates:** Set to download but not auto-install.
11. **NIC Configuration:** Team network cards, set full duplex (e.g., 1 Gbps).

I/O & Monitoring

12. **Validate I/O Subsystem:** Test performance with tools like SQLIO.
13. **SAN Tuning:** Work with storage admins to optimize queue depth & MPIO.
14. **Anti-Virus Setup:** Install AV and point it to update server.
15. **Monitoring:** Add server to monitoring tools (e.g., SCOM).

🔧 **2. Pre-Installation of SQL Server**

Drive Preparation

1. **Separate RAID Arrays:**
    - Data

- Logs
- TempDB (dedicated)

2. Format Data, Log, and TempDB drives with **64K allocation unit size** for SQL performance.

Service Accounts

3. Add SQL Server Admins group to local Administrators.
4. Create Active Directory service accounts for SQL services; grant proper SPN permissions.

Drive Letter Standards

5. Set drive letters consistently (e.g., E: for Data, L: for Log, T: for TempDB).

Folder Structure

6. Create and secure folders for SQLData, SQLLogs, and TempDB before install, granting full control to service accounts.

🛠 3. SQL Server Installation Steps

Install SQL Server

1. Use the **pre-configured service accounts** for SQL services.
2. Install binaries to **D: or another non-C drive** for better separation.
3. Set SQL Server & SQL Agent to **Automatic start**. Disable SQL Browser unless needed.
4. Apply latest **Service Pack and cumulative updates** after install.
5. Add SQLAdmins group to **sysadmin** fixed server role.

🧹 4. Post-Installation Configuration

Security & Permissions

1. Grant SQL Server service accounts full control on Data/Logs/TempDB folders.
2. Remove individual service accounts from root folders — rely on service group permissions.

Local Policies

3. Add service group accounts to Windows privileges such as **Lock Pages in Memory** and **Perform Volume Maintenance Tasks**.

Anti-Virus Exclusions

4. Exclude SQL Data, Log, TempDB, backup paths, and binaries from AV scanning.

SQL Security

5. Remove built-in Admins from sysadmin role; enable failed login auditing.
6. Enable **TCP/IP** and change default port from 1433 for security.

Server & Agent Features

7. Enable **remote DAC** for emergency troubleshooting.
8. Enable features like **DatabaseMail**, **SQLCLR**, and **xp_cmdshell** only if required and secure, and configure proxy accounts if used.

Logging & Jobs

9. Configure SQL error log retention (e.g., 30 files).
10. Create SQL Agent jobs for:

- Backups
- Index maintenance
- Stats updates
- Cleanup jobs (backup & history)

Database Configuration

11. Move **MSDB** database files to Data/Log folders.
12. Reconfigure **TempDB** for number of data files based on CPUs and size them evenly with sensible autogrowth.
13. Enable trace flags (e.g., **1118** on older versions) for TempDB contention reduction.
14. Set **Model** database defaults (e.g., SIMPLE recovery by default) to ensure new databases have sane defaults.

Memory & Parallelism

15. Configure **max server memory** based on installed RAM, leaving room for OS and other processes.
16. Set **max degree of parallelism (MaxDOP)** appropriate to workload and cores.
17. Set **cost threshold for parallelism** to a more suitable value (e.g., 20–25+ for OLTP workloads).

Administrative Best Practices

18. Create an AD login for emergency access (patching/troubleshooting).
19. Standardize the **SA account password** and rotate regularly.
20. Ultimately, remove SQL Admins from local Administrators for least privilege.

🧠 **Summary**

This checklist ensures that SQL Server installations are:

✓ Secure

✓ Consistent

✓ Performance-oriented

✓ Maintainable & supportable over time.

SQL Server Installation Checklist

1. Windows / OS Preparation

☐ OS installed on **C:\ drive**

☐ Latest Windows patches applied

☐ Hyper-Threading enabled in BIOS

☐ Power Plan set to **High Performance**

☐ Partition alignment verified

☐ Windows Updates set to *Download only*

☐ Server added to monitoring system

☐ Antivirus installed and configured

☐ SQL folders excluded from antivirus scanning

☐ Network NICs configured (teaming, full duplex)

2. Disk & Storage Configuration

☐ Separate disks created for:

- ☐ Data
- ☐ Logs
- ☐ TempDB
- ☐ Backups

☐ Data, Log, TempDB disks formatted with **64K allocation unit size**

☐ Drive letters standardized (example):

- ☐ D:\Data
- ☐ L:\Log
- ☐ T:\TempDB
- ☐ B:\Backup
- ☐ S:\SystemDB

☐ I/O subsystem tested (latency & throughput)

3. Security & Accounts

☐ Domain service accounts created for:

- ☐ SQL Server Engine
- ☐ SQL Server Agent

☐ Service accounts granted:

- ☐ Perform Volume Maintenance Tasks (IFI)
- ☐ Lock Pages in Memory

☐ SQL Admins AD group created

☐ SQL Admins added to local Administrators

☐ Strong password & account lockout policies enforced

4. SQL Server Installation

☐ SQL binaries installed on **non-C drive**

☐ Latest SQL Server CU installed

☐ SQL Server service startup = Automatic

☐ SQL Agent startup = Automatic

☐ SQL Browser disabled (if not required)

☐ SQL Admins group added to sysadmin role

5. System Databases Configuration

☐ Master database moved to **SystemDB drive**

☐ Model database configured with standard defaults

☐ MSDB database moved to Data/Log drives

☐ System DB backups configured

6. Post-Installation Configuration

☐ Max Server Memory configured

☐ MaxDOP configured

☐ Cost Threshold for Parallelism configured

☐ Instant File Initialization enabled

☐ Optimize for Ad Hoc Workloads enabled

☐ Remote DAC enabled

7. TempDB Configuration

☐ TempDB placed on dedicated drive

☐ Number of TempDB data files set (CPU-based)

☐ All TempDB data files same size

☐ Fixed autogrowth configured

☐ TempDB pre-sized

8. Maintenance & Jobs

☐ Backup jobs created (Full, Diff, Log)

☐ Index maintenance jobs created

☐ Statistics update jobs created

☐ Cleanup jobs configured

☐ SQL error log retention configured

9. Security Hardening

☐ SA password changed and secured

☐ Built-in Administrators removed from sysadmin

☐ Failed login auditing enabled

☐ TCP/IP enabled

☐ SQL port configured (non-default if required)

✅ 10. Final Validation

☐ Test connection to SQL Server

☐ Validate backup & restore

☐ Verify monitoring alerts

☐ Document server configuration

✔ **Installation Completed By: ____________________**

✔ **Date: ____________________**

SQL Server Installation Checklist *(Azure / AWS / Always On)*

OS / VM PREPARATION

☐ OS installed and fully patched

☐ Power Plan set to **High Performance**

☐ Time zone and NTP configured

☐ Azure VM size / AWS EC2 instance type validated

☐ Accelerated Networking enabled (Azure / AWS)

☐ Windows Defender / AV exclusions planned

STORAGE CONFIGURATION

☐ Separate disks created for:

- ☐ Data
- ☐ Logs
- ☐ TempDB
- ☐ Backups

☐ Premium SSD (Azure) / io2 or gp3 (AWS) used

☐ Disk caching configured correctly

- Data → ReadOnly
- Logs → None
- TempDB → None

☐ NTFS/ReFS formatted with **64K allocation unit size**

☐ I/O latency tested and validated

SECURITY & SERVICE ACCOUNTS

☐ Domain service accounts created

☐ SQL Admins AD group created

☐ SQL Admins added to local Administrators

☐ Instant File Initialization enabled

☐ Lock Pages in Memory enabled

☐ Least privilege model applied

SQL SERVER INSTALLATION

☐ SQL binaries installed on non-OS drive

☐ Latest CU applied

☐ SQL Server service = Automatic

☐ SQL Agent service = Automatic

☐ SQL Browser disabled (if not required)

☐ SQL Admins group added to sysadmin

SYSTEM DATABASES

☐ Master database moved from C:\

☐ Model database configured with standard defaults

☐ MSDB moved to Data/Log drives

☐ System DB backups configured

TEMPDB CONFIGURATION

☐ TempDB on dedicated disk

☐ Multiple TempDB data files (CPU-based)

☐ All data files same size

☐ Fixed autogrowth configured

☐ TempDB pre-sized

POST-INSTALL SQL CONFIGURATION

☐ Max Server Memory configured

☐ MaxDOP configured

☐ Cost Threshold for Parallelism set

☐ Optimize for Ad Hoc Workloads enabled

☐ Remote DAC enabled

ALWAYS ON AVAILABILITY GROUP (IF APPLICABLE)

☐ Windows Failover Cluster validated

☐ Quorum configured correctly

☐ Availability Group created

☐ Listener configured and tested

☐ Backup preference configured

☐ Read-only routing tested

MAINTENANCE & MONITORING

☐ Full, Diff, Log backups configured

☐ Index maintenance jobs created

☐ Statistics maintenance jobs created

☐ Cleanup jobs configured

☐ Monitoring & alerts enabled

✅ **FINAL VALIDATION**

☐ Application connectivity tested

☐ Backup & restore tested

☐ Failover tested (if AG)

☐ Configuration documented

Installed By: ______________________

Date: ______________________